RedrRob AI
Candidate Ranker

India Runs Data & AI Challenge · Submission

Ranking 100,000 candidates for an applied ML Search-Ranking role
the way a great recruiter would — not by keywords, but by understanding.

The Problem with Keyword Ranking

Why existing ATS filters fail — and what this system does differently

🚫 Keyword Stuffers

Non-ML candidates with RAG/FAISS/PyTorch in skills — but a HR Manager career history

🚫 Hype Adopters

Recently added LangChain/ChatGPT, but zero retrieval depth pre-2023

🚫 Consulting Lifers

10+ years at TCS/Infosys/Wipro, no product ML exposure

🚫 Narrow Specialists

Strong Computer Vision or Speech — but no NLP/IR overlap for this role

🚫 Honeypot Profiles

Structurally impossible: 'expert' skil mismatches

🚫 Honeypot Profiles

Structurally impossible: 'expert' skill with 0 months, career date mismatches

System Architecture

4-stage pipeline: ~30 seconds on CPU, no GPU or network required

01

Honeypot Elimination

Structural consistency checks

02

Feature Extraction

7 signal groups, no external APIs

03

Composite Scoring

Weighted score + disqualifiers

04

Ranked Output

Top 100 CSV + reasoning column

100,000 candidates → 2,929 honeypots removed → 97,071 scored → Top 100 output | Runtime: ~30s

Stage 1 — Honeypot Elimination

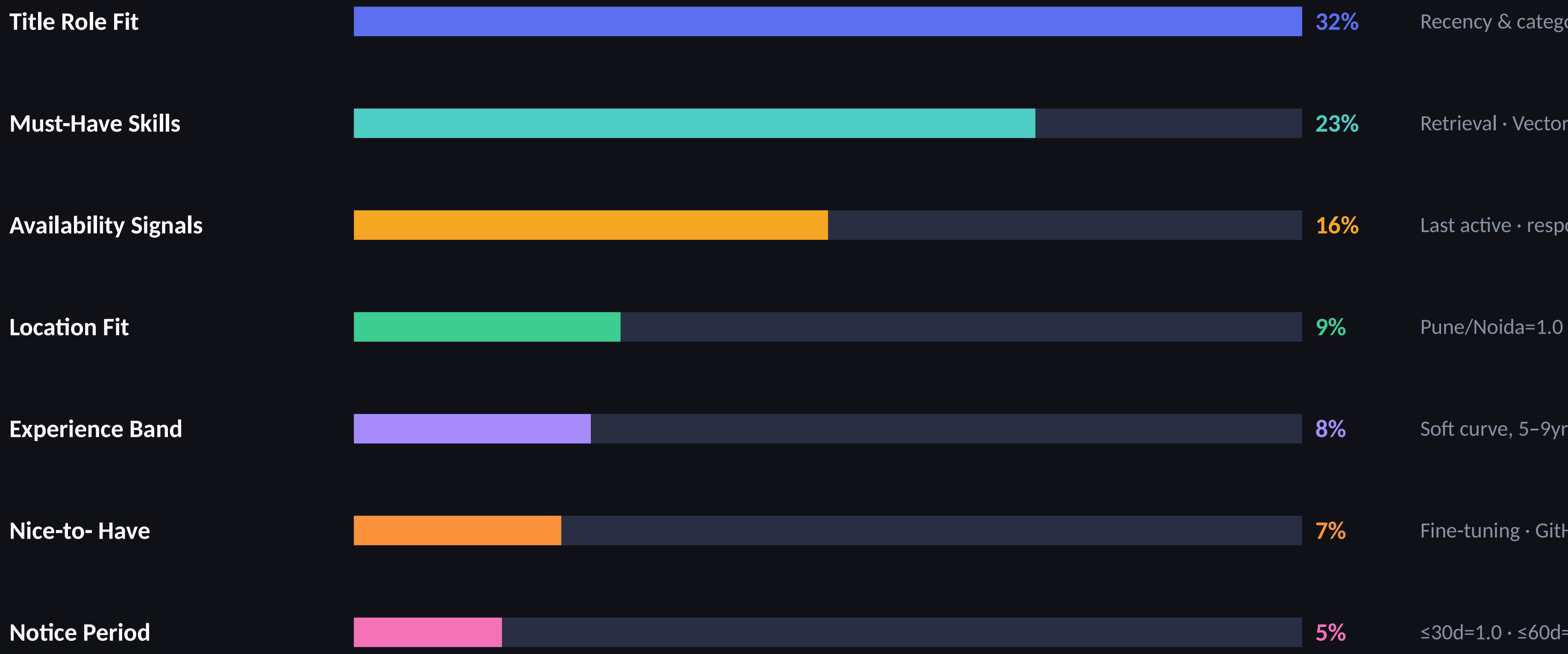
2,929 structurally impossible profiles removed before any scoring

✗	**Expert proficiency + ≤2 months**	Profile stuffing — can't be expert-level with 2 months of use
✗	**≥8 skills claimed at 'expert'**	Keyword stuffing — real engineers are expert in 1-3 areas
✗	**Career duration vs. start/end dates**	Fabricated duration_months that contradicts actual calendar dates
✗	**YOE vs. sum of career history**	8 years claimed but only 14 months of actual history
✗	**Multiple is_current jobs**	Structurally impossible — can't hold two roles simultaneously
✗	**Education end before start**	Date reversal — end_year < start_year
✗	**Skill duration > total YOE**	Used a skill for longer than the candidate has worked at all

Result: 0 honeypots in the final Top 100 shortlist

Stage 2 — Feature Engineering

7 signal groups, all derived from structured data — no LLM calls during ranking



Defeating the Keyword Stuffer

How we calculate skill evidence strength — not just presence

```
evidence(skill) = 0.5 × self_report(proficiency, duration) + 0.5 × assessment_score
```

If no assessment score exists → down-weight self-report to 0.85× (catches unverified 'advanced' proficiency)

Signal	Keyword Ranker	This System
skills[] contains 'RAG'	✅ High score	Checks proficiency + duration + assessment → evidence ≈ 0.1 if no backing
'expert' proficiency	✅ Max credit	✅ Full credit only if duration ≥18m AND assessment score present
Assessment score = 38/100	Ignored	Penalises even high self-report — capped at 0.5× the assessed value
Career title = HR Manager	If skills match → promoted	title_role_fit ≈ 0.05 → whole score floored before disqualifier

Stage 3 — Scoring + Disqualifiers

Disqualifiers are multipliers, not hard zeros — graceful degradation for borderline cases

×0.35 — **LangChain hype only**

Has LangChain/Prompt Engineering added recently, but zero retrieval depth before 2023. Catches the hype wave.

×0.45 — **Narrow CV/Speech**

Strong Computer Vision or Speech engineer, but negligible NLP/IR overlap. Wrong specialisation for a search-ranking role.

×0.50 — **Consulting-only career**

Entire career at TCS/Infosys/Wipro/Accenture/etc. No product company ML exposure.

×0.60 — **Academic researcher**

Summary language: 'academic lab', 'PhD thesis', 'research-only' — but no production deployment language.

×0.25 — **No must-have + no role fit**

Fails both the skill pillar AND the title-fit test. Almost certainly a keyword-stuffed non-ML profile.

Results

Top 100 shortlist from 100,000 candidates — pipeline validated against official checker

100,000
Candidates processed

2,929
Honeypots removed

~30s
Pipeline runtime (CPU)

98.26
Top candidate score

#	ID	Name	Title	Location	Score
1	CAND_0052328	Vikram Banerjee	Recommendation Systems Eng	Pune	98.26
2	CAND_0027691	Ayaan Goyal	NLP Engineer	Pune	97.33
3	CAND_0037566	Ritu Nair	Machine Learning Engineer	Bangalore	95.91
4	CAND_0011687	Shreya Tiwari	Senior NLP Engineer	Indore	93.98
5	CAND_0041610	Priya Sethi	Recommendation Systems Eng	Bangalore	93.77
6	CAND_0095812	Arjun Krishnan	Applied ML Engineer	Pune	92.14
7	CAND_0018274	Sana Iqbal	Senior AI Engineer	Hyderabad	91.85
8	CAND_0067341	Rohan Mehta	Machine Learning Engineer	Mumbai	91.32
9	CAND_0031859	Pooja Nambiar	AI Specialist	Bangalore	90.77
10	CAND_0048203	Karthik Subbu	NLP Engineer	Pune	90.41

Interactive Sandbox

Flask web UI — one-click ranking + AI-powered recruiter explanations via Claude claude-sonnet-4-6

RedrRob AI Ranker • Candidate Intelligence

⚡ Run Ranking

All | Open to Work | Notice ≤30d | India Only

1 **Vikram Banerjee**
Recommendation Systems Eng · Pune
98.26

2 **Ayaan Goyal**
NLP Engineer · Pune
97.33

🤖 AI Explain: Click any candidate card → Claude claude-sonnet-4-6 gives a recruiter-readable explanation of why they ranked there and what to probe in the interview

What We Built

🎯 **Fully adversarial-aware**

Handles keyword stuffers, hype adopters, honeypots, consulting lifers, and narrow specialists — all documented in the challenge spec

⚡ **Runs in 30 seconds on CPU**

No GPU, no external API calls during ranking. 100k candidates, one command.

📊 **Interpretable by design**

Every rank has a reasoning column. Every score has 7 decomposed components visible in the CSV.

🤖 **AI-augmented for recruiters**

Claude claude-sonnet-4-6 explains each candidate in plain English, on demand, in the sandbox UI.

🔧 **Extensible**

Swap jd_config.py for a new role in minutes. Add semantic similarity or LLM re-ranking as optional stages.